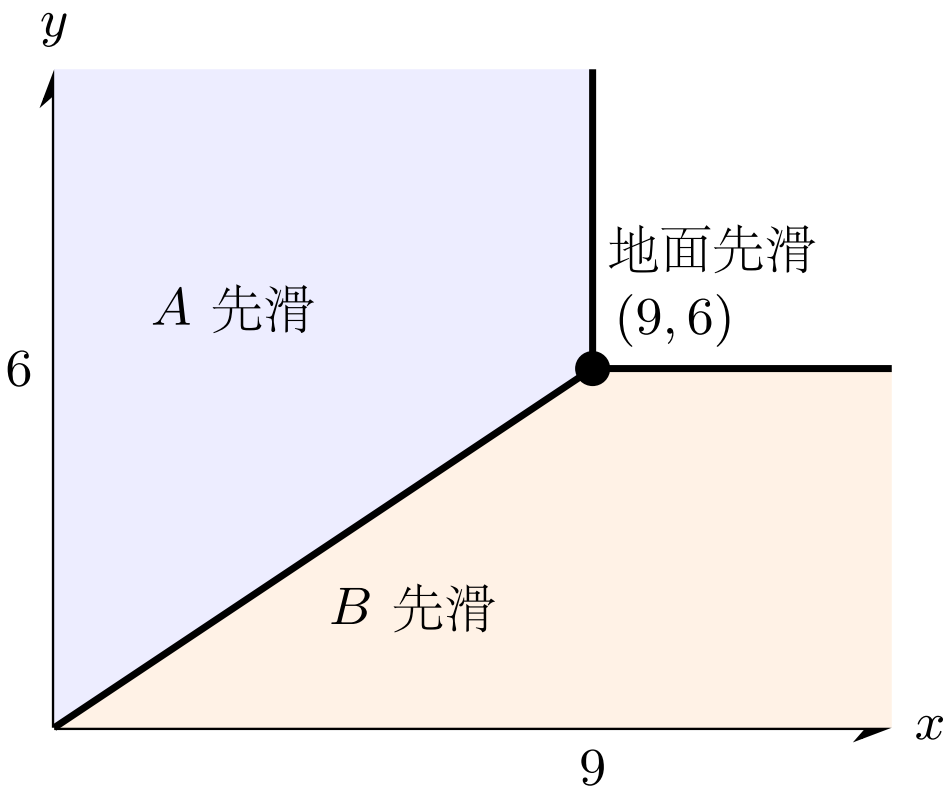
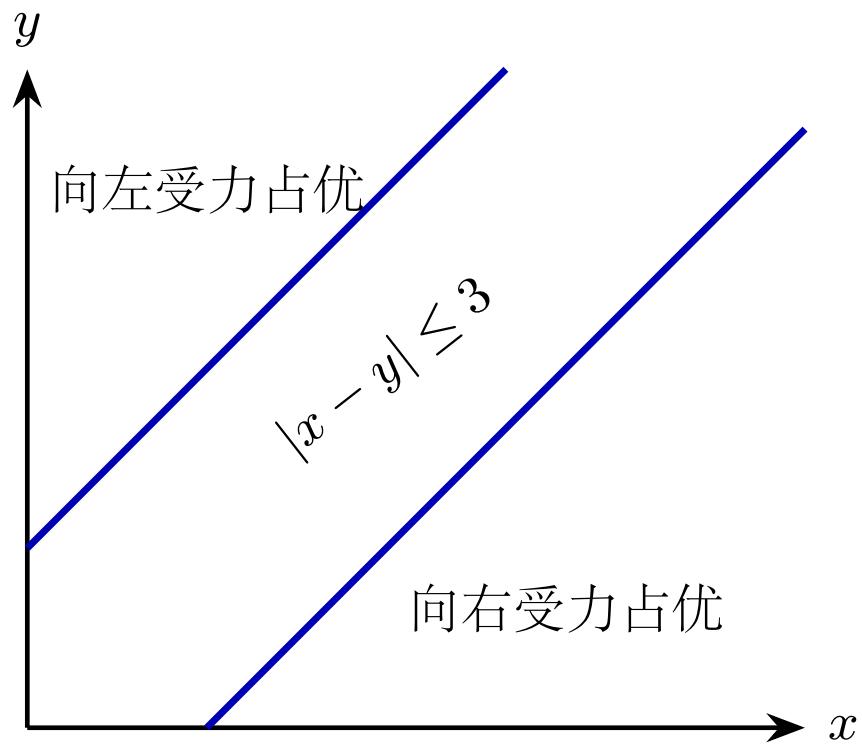


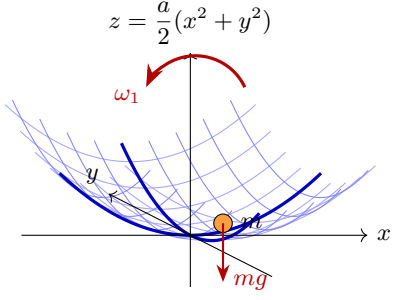
绳固定于 A 的下结点
绕 B 右半轮，再绕 A 左半轮
三股拉 A 向右，两股拉 B 向左



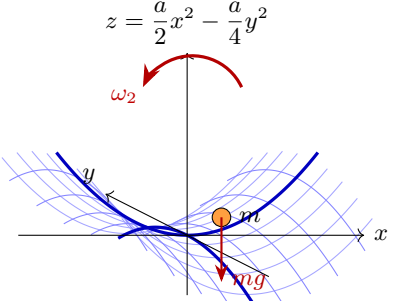


左：首次启动的有效边界段；右：上层两处均滑后的受力条件

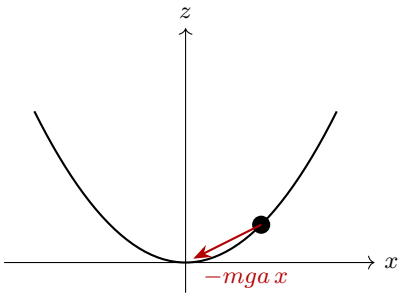
中带允许静止；已经运动的底块先减速，不能瞬时停下


$z = \frac{a}{2}(x^2 + y^2)$
ω_1
y
m
x
mg

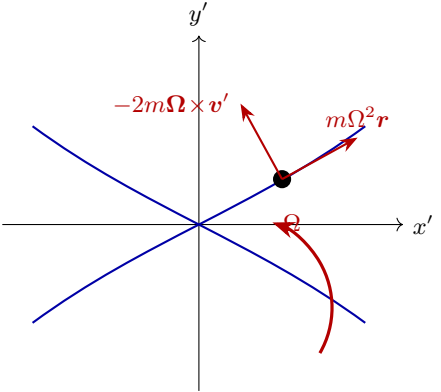
旋转抛物面


$z = \frac{a}{2}x^2 - \frac{a}{4}y^2$
ω_2
y
m
x
mg

旋转马鞍面



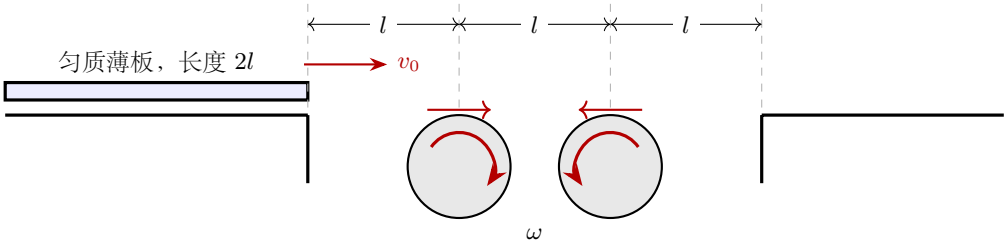
稳定抛物面：$\omega_0^2 = ag$

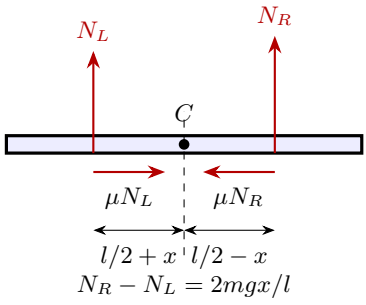


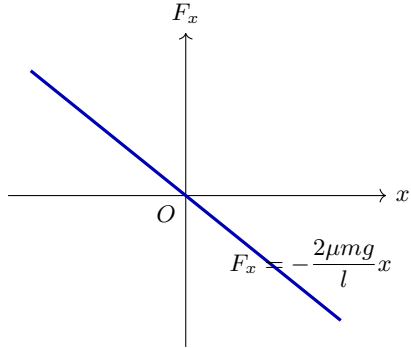
$$z = \frac{a}{2}x'^2 - \frac{a}{4}y'^2$$

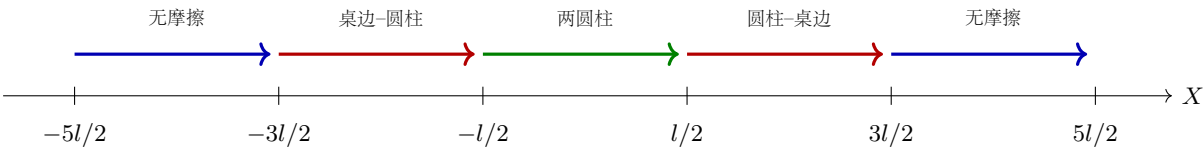
稳定下确界：$\Omega_c^2 = ag$

严格稳定 $\Omega > \sqrt{ga}$；等号允许 $x \propto t$


匀质薄板，长度 2l
v0
l
l
l
ω

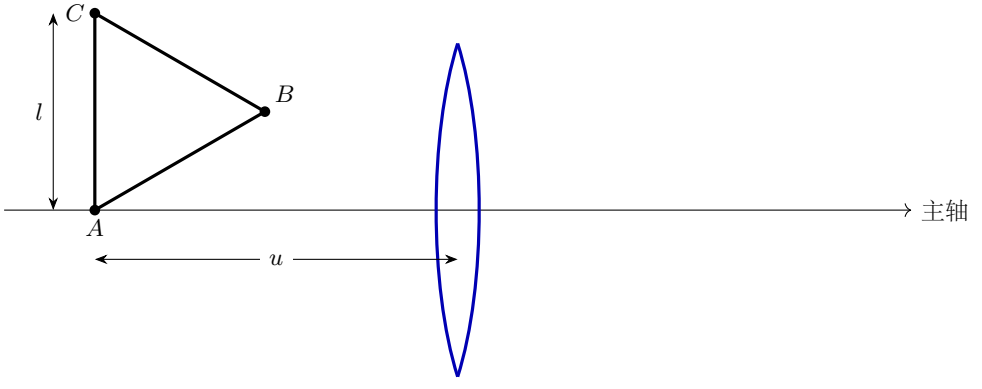




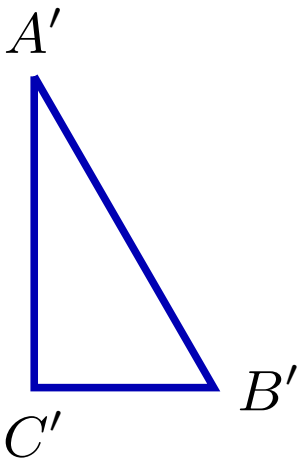


速度事件另分段：$v > u$、$0 < v < u$、$v = u$

$v = u$ 且 $X \le 0$ 可有静摩擦恒速平台

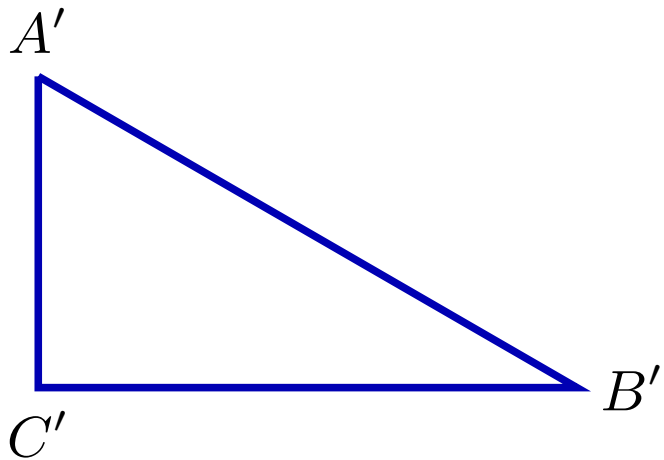

C
B
l
A
u
主轴
薄凸透镜

三组逐点薄透镜像坐标；同组横、纵比例相同



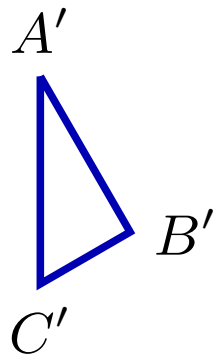
$u/f = 4$

$l/f = 1.7321$



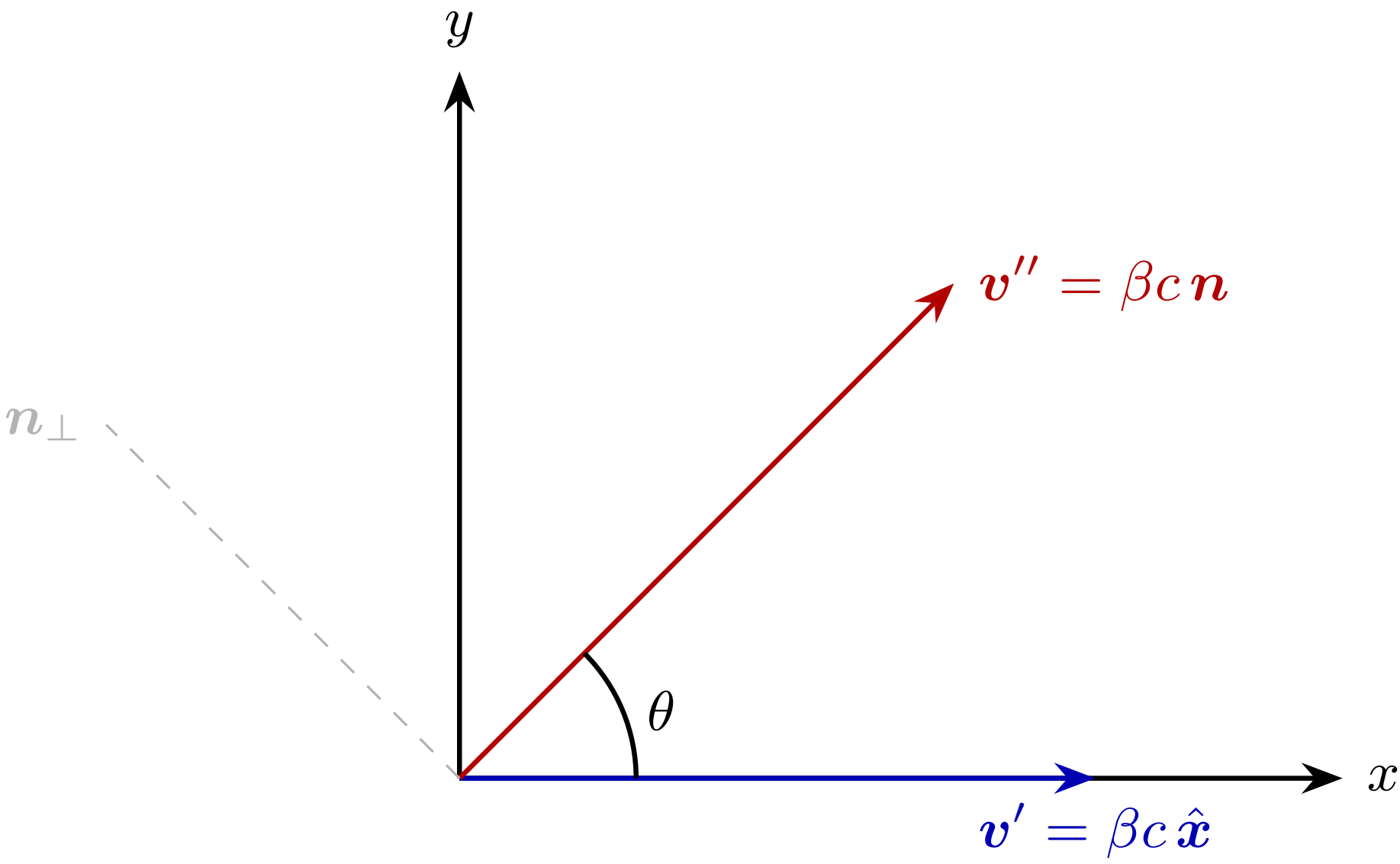
$u/f = 2$

$l/f = 0.5774$



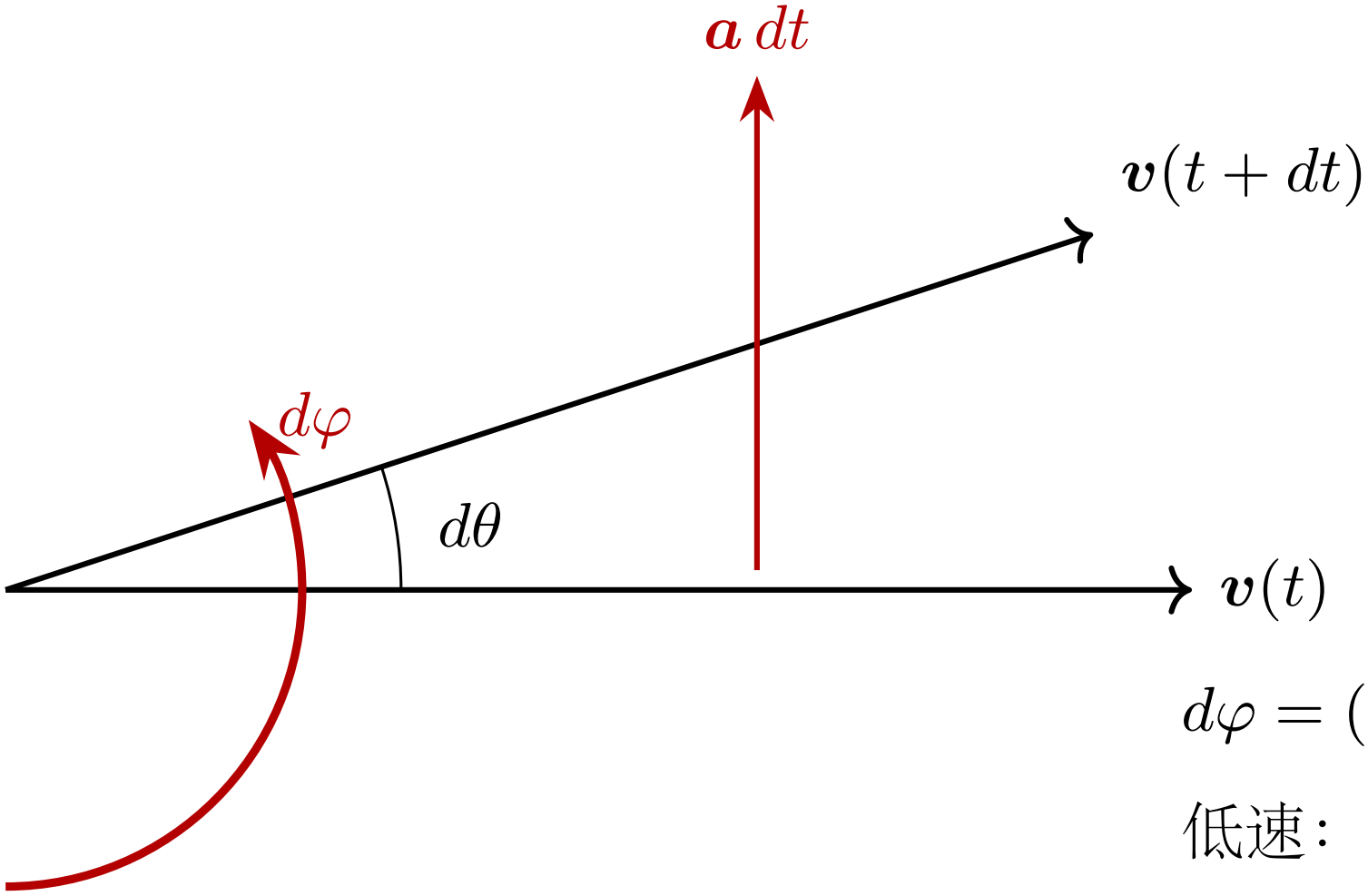
$u/f = 4$

$l/f = 1.1547$



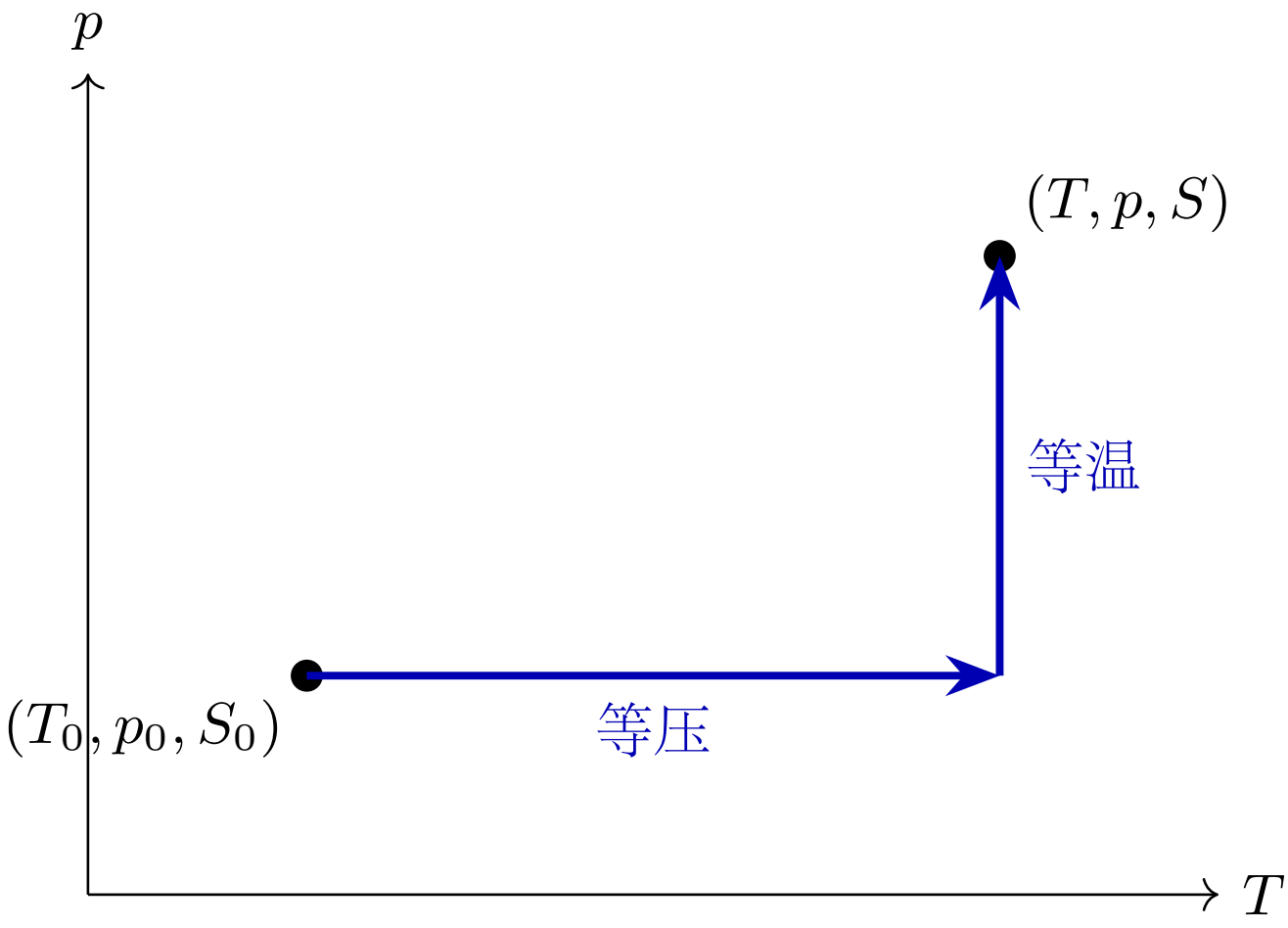
两速度均从 S 指定；先分别作标准洛伦兹推动

再复合 $\Lambda''(\Lambda')^{-1}$ 求相对变换



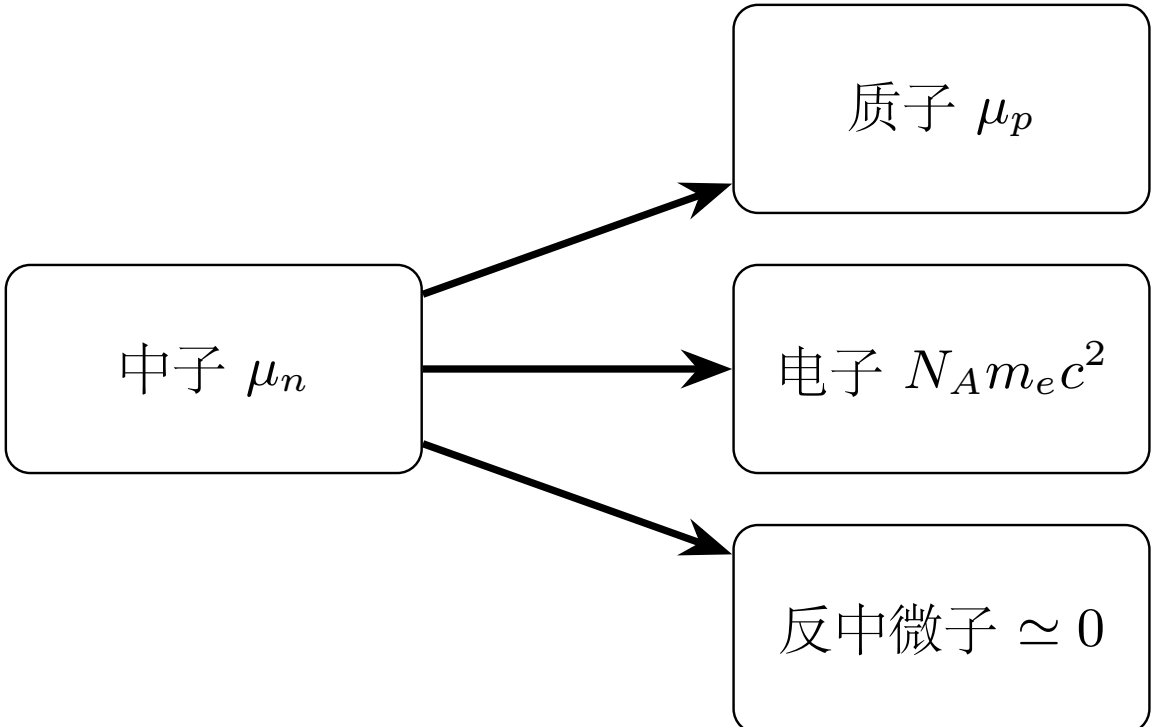
$$d\varphi = (\gamma - 1)d\theta$$

低速：$\boldsymbol{\omega}_t \simeq \dfrac{\boldsymbol{v} \times \boldsymbol{a}}{2c^2}$



$$n \rightleftharpoons p + e^- + \bar{\nu}$$

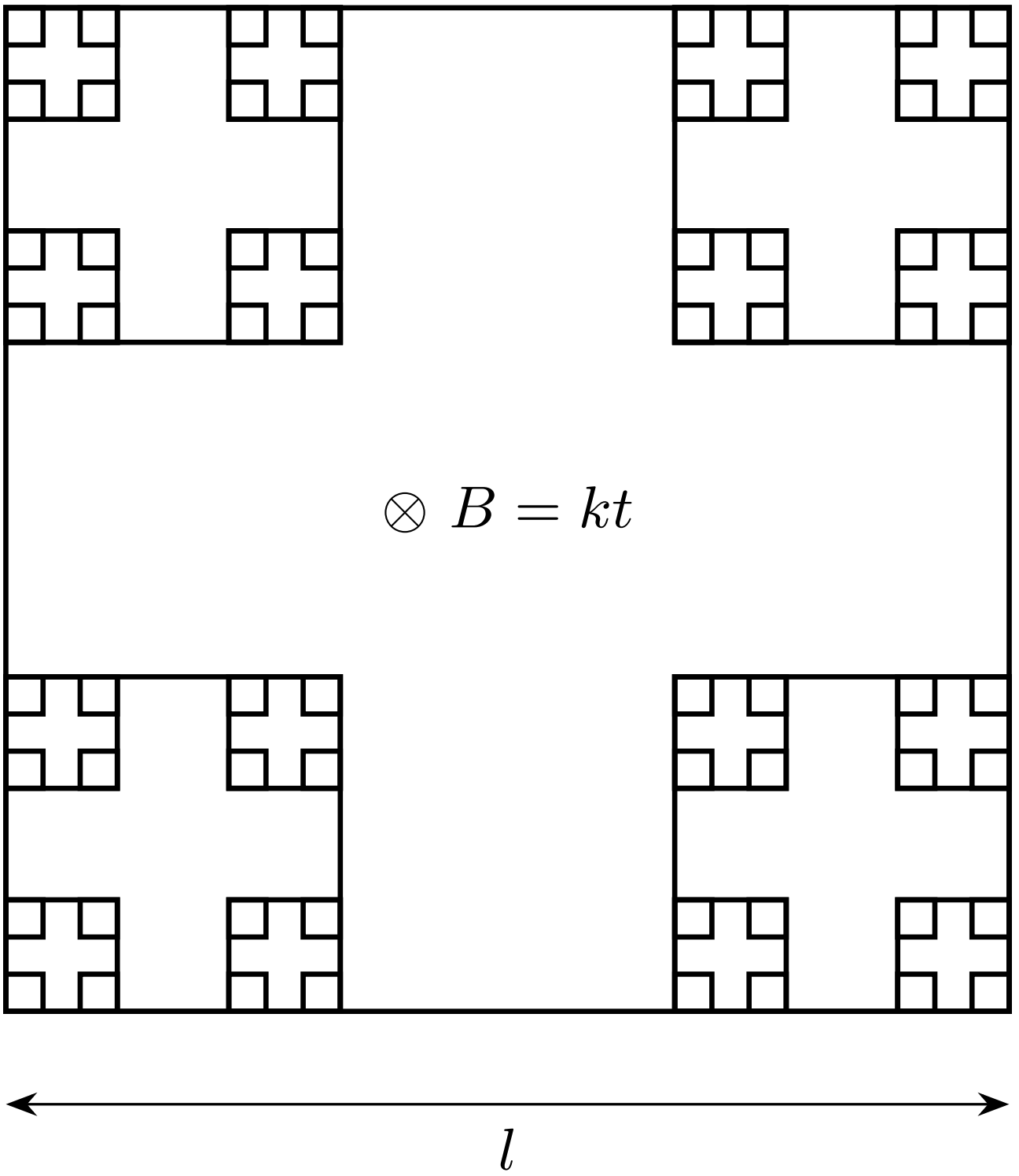
$$\mu_n = \mu_p + \mu_e + \mu_{\bar{\nu}}$$

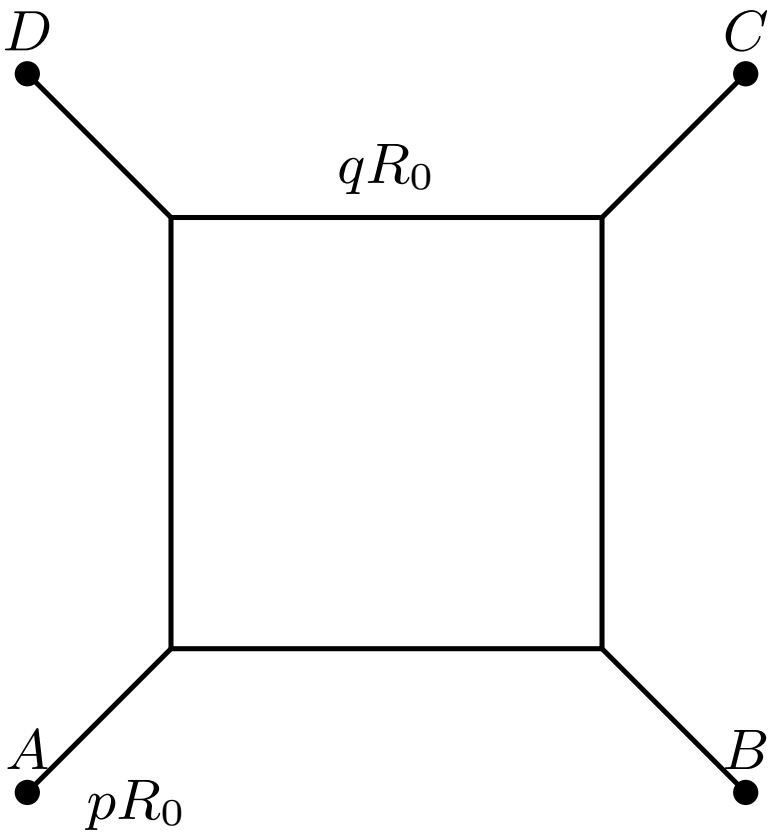


$$RT \ln \frac{1 - x_p}{x_p} = N_A m_e c^2$$

$$x_p = \frac{1}{1 + e^{N_A m_e c^2/(RT)}}$$

仅画有限层；重合边为同一导线

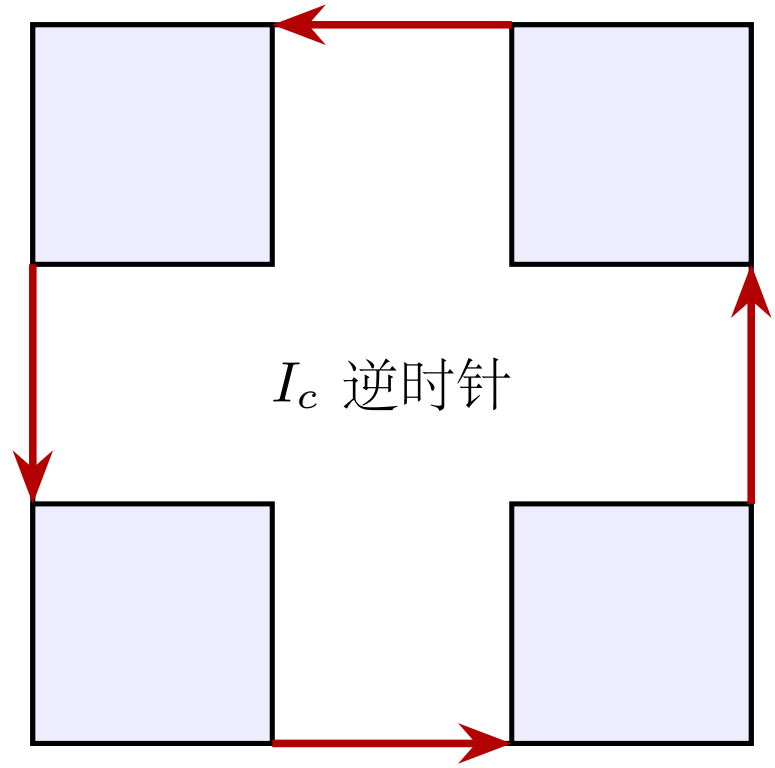




$$p = 2x - y = 1/14$$

$$q = 4(y - x) = 4/7$$

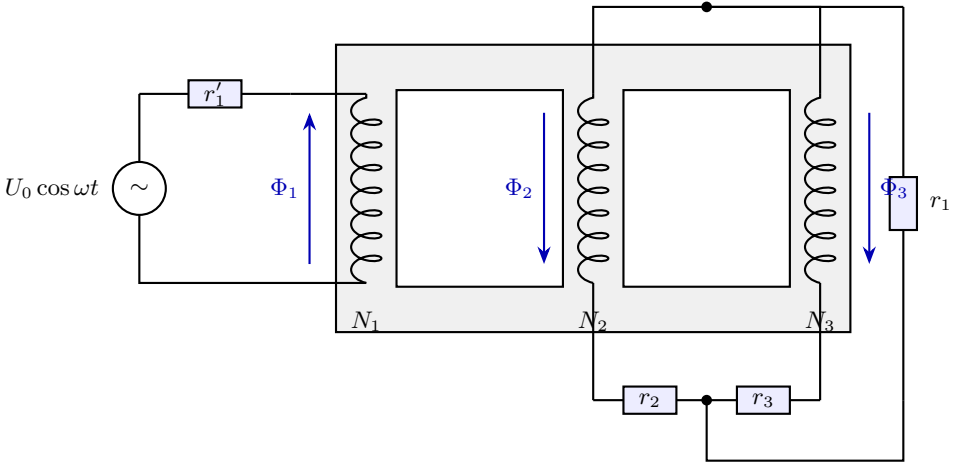
无源四端口等效



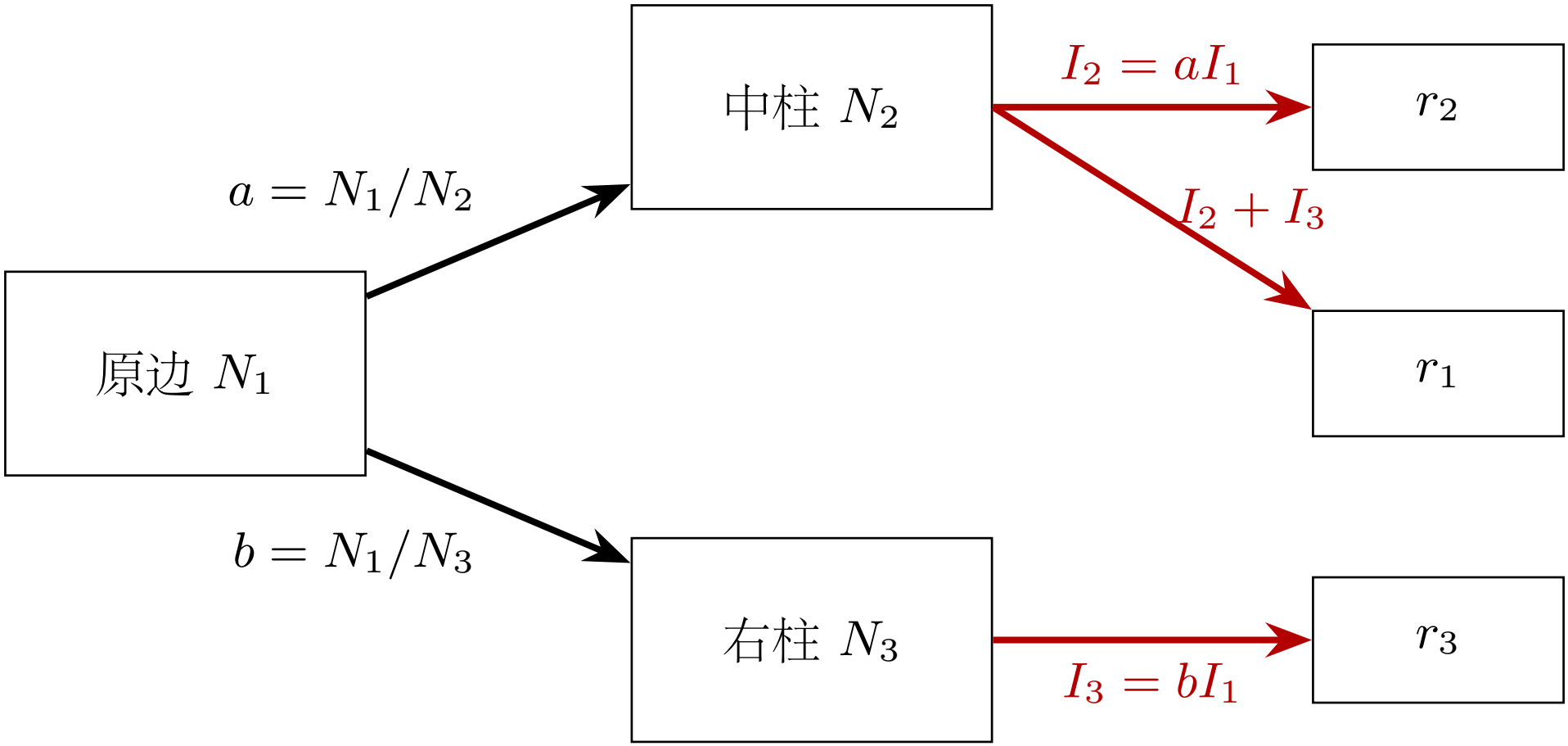
$$R_{\text{top}} = 16R_0/7$$

$$\mathcal{E}_{\text{eff}} = 7kl^2/9$$

每角子网络独立证明功率正交



三连续绕组、次级上端共结点；下端经 r_2, r_3 汇入 r_1



$$D = r_1' + a^2 r_2 + b^2 r_3 + (a+b)^2 r_1$$

$$P_2 = \frac{1}{2} r_2 a^2 \left(\frac{U_0}{D} \right)^2$$